Post-2020 Inflation Acceleration Signals Heightened systemic Pressure in Global Food Markets.

Average Inflation (%)

14.69

Average Volatility (Price Range)

0.09

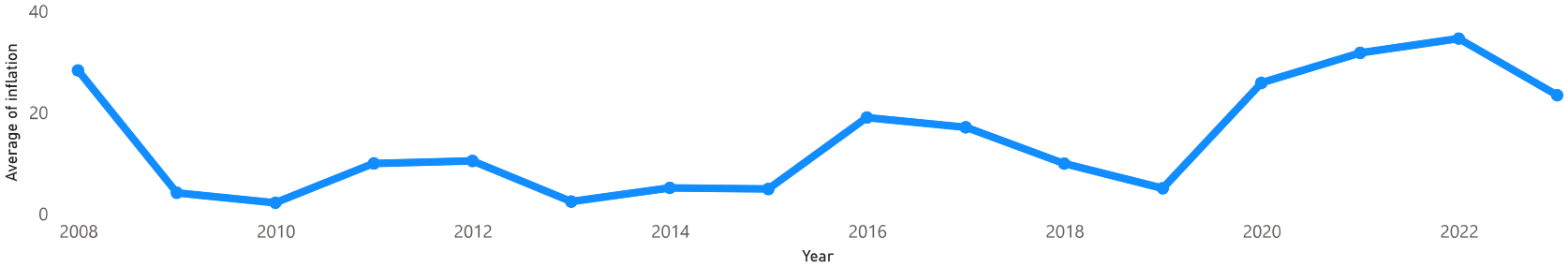
Country Count

25

High Inflation (95th percentile)

68.15

Global Average Food Inflation (2007-2023)



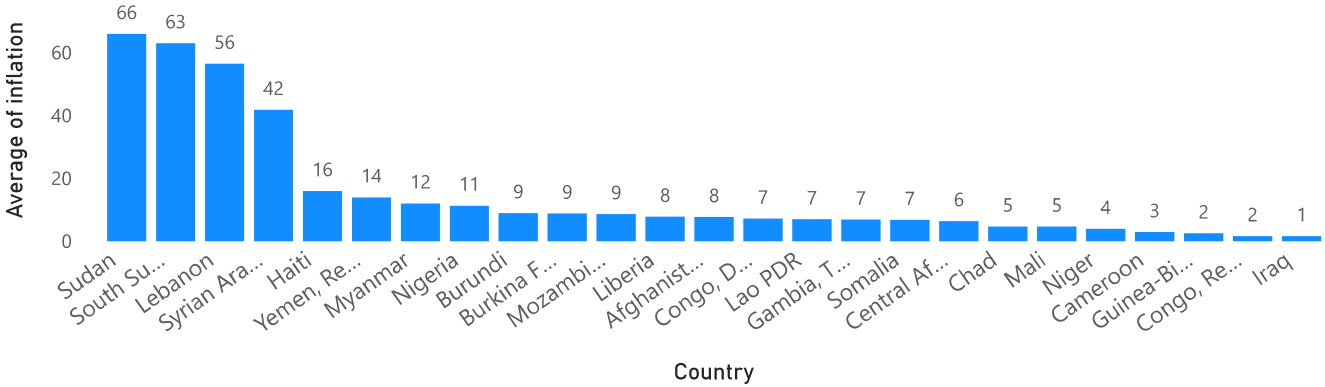
country

- Afghanistan
- Burkina Faso
- Burundi
- Cameroon
- Central Africa...
- Chad
- Congo, Dem. ...
- Congo, Rep.

The post-2020 inflation surge indicates heightened systemic risk in global food markets. Policymakers and procurement teams should prioritise volatility monitoring and price stabilisation strategies.

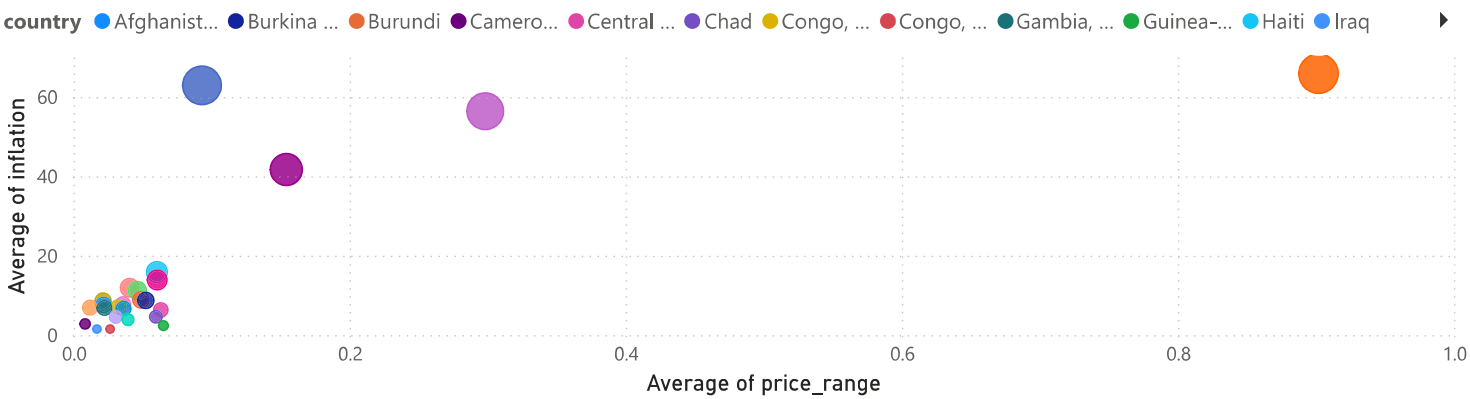
Structural Drivers of Food Price Inflation

Average Food Inflation by Country



Statistical testing confirms significant regional differences in inflation levels (Kruskal-Wallis, $p < 0.05$). Some countries consistently experience higher food price pressure.

Volatility and Inflation Relationship



Correlation

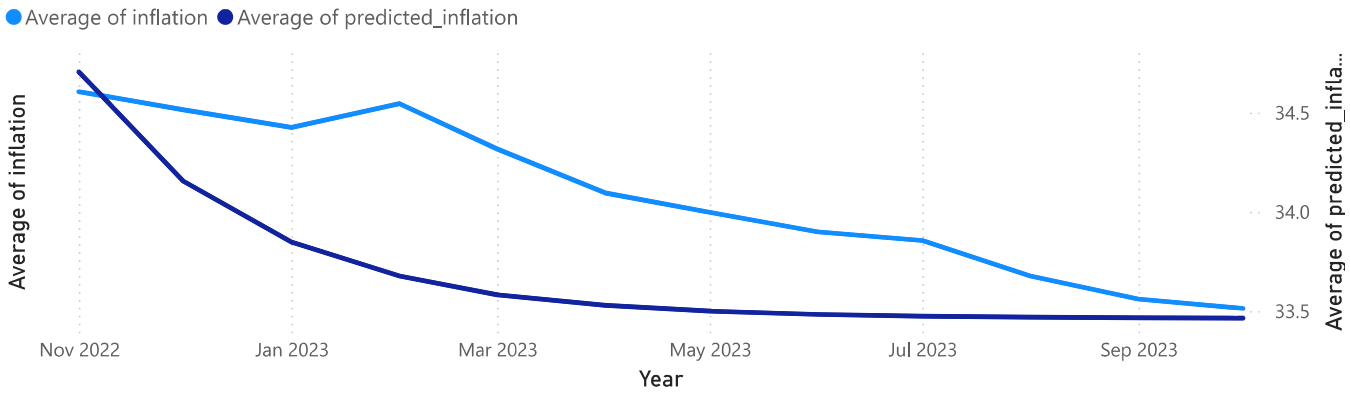
0.31

A weak positive correlation ($r = 0.31$) suggests that higher price volatility is associated with higher inflation, though volatility alone does not strongly explain inflation differences across countries.

Model Captures Directional Trend with Moderate Accuracy

MAE	RMSE
8.93	10.29

Actual vs Forecasted Inflation (12-Month Holdout)



The forecasting model captures the general downward trend in inflation during the holdout period. However, it underestimates the magnitude of decline in later months, suggesting moderate predictive performance suitable for short-term monitoring rather than precise forecasting.